PROJECT GAP AUDIT — ATMOSphere, measured 2026-07-23 (Phase 5a of BG-QUALITY-ROOTCAUSE)

Field	Value
Revi-sion	1
Cre-ated	2026-07-23
Last modi-fied	2026-07-23T04:59:28Z
Status	COMPLETE — read-only audit; every count measured this session with its command (Appendix A); §11.4.201(7)(b) control needles before every reported absence
Author	(T1/main - claude1) Phase-5a audit subagent, Fable, §11.4.182
Inputs	`../ROOT_CAUSE_ANALYSIS.md` (PC-1..PC-9, MU-1..MU-6, M1..M19) · `../OPERATOR_COMPLAINT_MINING.md` (+ `../complaints.tsv`) · `../solutions/README.md` + SOL-01..SOL-10
Man-date	Operator directive (verbatim): *"Current project MUST BE fully evaluated and fixed after everything is fully done and all violations, gaps and weak spots and danger zones fixed! We MUST make sure that next 0.0.9 release that we will ship DOES WORK from 'a to z' and for QA team during manual testing to find any bug/issue or inconsistency to be almost impossible!"*
Con-sumer	Phase 5b remediation (GATED on the Phase-4 `anti_bluff` submodule for its helper layer; per-item gating stated in §4)

. The honest bar (§ . .)

"Almost impossible for QA to find a bug" is achievable as: **(a) the known-recurring failure classes made mechanically impossible to ship** (a terminal status cannot exist without class-matched evidence; a release tag cannot mint while any registered guard lacks a verdict on the candidate artifact; a recurrence cannot silently re-enter as a fresh id), **plus (b) real detection pressure applied BEFORE QA** (the first-touch smoke + the most-reopened set exercised on the flashed candidate, first). It is NOT a guarantee of zero defects — no mechanism below claims that, and any document that did would itself be the §11.4 bluff this audit exists to prevent. The measured discriminator (ROOT_CAUSE §7) is the design target: every fix whose done-claim was preceded by an on-device RED→GREEN flip on the shipping artifact recorded zero reopens; every fix confirmed at source-green bounced. Phase 5b's job is to make that discriminator the only path to "done" — and to run it against the 0.0.9 candidate before QA touches it.

. Executive summary — the gap register, counted

All figures measured 2026-07-23 in this checkout (commands in Appendix A). Where a figure reproduces a ROOT_CAUSE baseline it is flagged `[=M#]`; deltas are stated.

#	Gap category	Measured magnitude (today)	Closes with	First-touch-QA risk
G1	Status-without-custody (PC-1)	52/108 terminal items zero-history (48%) `[=M2]` ; 13/26 Reopened without a Reopened event `[=M4]` ; **14 items** with impossible sequences (reopens>0, terminal events=0); **173/187 (92.5%) done-claimers unguarded** `[=M7]` ; 25/25 reopen events "AI" `[=M6]`	SOL-01 + SOL-04	HIGH — this is the population the operator's 6/6-broken sample came from
G2	Unimplemented named gates (PC-4)	**239/413** CM-* name-literals with no implementation in the two canonical gate sites (upper bound — see the measured join-gap in §2.2); 85 explicit "gate-code = separate work item" deferrals `[=M12]` ; custody seams (sweep, ratchet, `CM-STATUS-CUSTODY`) still absent `[=M11]`	SOL-05 (ledger + ratchet)	MEDIUM directly; HIGH indirectly (it is why G1/G6 persist)
G3	Tracker scope-coverage (SOL-08 axis)	**289/1173 (24.6%)** main-repo commits in the last 60 days carry a tracked id; 75.4% of commit volume runs beside the tracker	SOL-08	MEDIUM — untracked work is invisible to every custody/coverage seam

#	Gap category	Measured magnitude (today)	Closes with	First-touch-QA risk
G4	Operator-experienced recurrences invisible to the tracker (PC-6)	≥3 same-defect re-mint pairs with **zero cross-links in the DB** (ATM-812≙ATM-788, ATM-793≙ATM-352-family, ATM-800≙ATM-351-family — link queries returned 0); 6 operator-experienced recurrences with zero reopen events (mining §4); `duplicate-of` used only 5× ever	SOL-07 (+ SOL-01 intake)	**HIGHEST** — these are the defects that keep coming back and the ranking signal is blind to them
G5	Measurement/ FAIL-bluff surface (PC-5)	**60 files / 413 piped-`grep -q`-under-pipefail sites** (SOL-03 census re-confirmed; −1 site vs the 414 baseline); release-gating concentration: `pre_build_verification.sh` **118 sites**, `build.sh` 16, `regression_guard_suite.sh` 4, `commit_all.sh` 4, `release_tag.sh` 1	SOL-03	HIGH for release-blocking false-FAILs; the false-PASS direction is the worse polarity
G6	Cross-layer verification gaps (PC-3 / §11.4.108)	82 PENDING-BUILD claim-moves in 60 days; terminal-event evidence classes: **30/51 source-ish vs 21/51 runtime-ish**; live instance: **ATM-799 closed `Fixed` 07-18 with evidence literally "(PENDING-DEVICE)"**; verdict store holds **zero verdicts for the 0.0.8 artifact** (built 07-21 02:44; only snapshot = 07-19); that snapshot: 74 guard rows → **24 PASS / 26 FAIL / 24 SKIP**	SOL-02 + SOL-04 + a verdict run on the candidate	**HIGHEST** — the "bounces on first touch" set is exactly this

#	Gap category	Measured magnitude (today)	Closes with	First-touch-QA risk
G7	0.0.9 first-touch danger zones	61 open critical-severity items; ranked family list in §3 (wrong-display routing ≥9 ids incl. the in-flight ATM-816 refactor; audio switching ≥7 ids; subtitles 3-cycle 49-day chain; brightness pair with **zero guards**; 0.0.7's crash-on-boot = 0-minutes-to-first-defect precedent)	SOL-02/04 + targeted guards + §11.4.132 ordering	This IS the QA session

What is already fixed (measured, credit where landed): the release seam is genuinely wired — `scripts/testing/release_tag.sh:125-135` invokes `scripts/testing/release_verdict_coverage.sh`, and `regression_guard_suite.sh:577-586` exits 3 on any PENDING ("ABSENCE of a verdict blocks"). A first-touch smoke guard exists and is registered (`ATM-826` → `smoke_first_touch.sh`, 48 KB, executable, wired in `test_all_fixes.sh`, finding_ids covering ATM-809/810/811/814/819/821). The 2026-07-18 remediation of M9/M10 is real. The remaining holes are upstream of that seam (status-write, intake, measurement) and downstream of it (no verdict run has ever been minted on a release candidate — §5 UNKNOWN-6 of ROOT_CAUSE remains true today).

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. The measured record per gap

. G — Status-without-custody (PC-), on today's DB

DB: `docs/workable_items.db`, opened `sqlite3 -readonly`. Corpus needle: 619 distinct items (non-zero → path sighted) `[=M1]`.

- **Terminal-status items:** 108 (`Fixed` 50 + `Completed` 53 + `Implemented` 5). Of these, **52 have ZERO `item_history` rows of any kind** (48%). Needle: `item_history` holds 676 rows.

- **Terminal events vs terminal statuses:** only 51 terminal events exist (Fixed 24 + Implemented 2 + Completed 25) for 108 terminal-status items — even among the items WITH history, most terminal transitions left no terminal event.
- **Reopened without custody:** 13 of 26 currently-Reopened items have no `Reopened` event `[=M4]`.
- **Impossible sequences (the R1 class):** 14 items carry reopen events but ZERO terminal events — you cannot reopen what was never closed; the terminal transitions bypassed the ledger: `ATM-277 (4/0)` — R1 itself, the stale-paused-frame-on-2nd-display defect, still `Reopened` today — plus `ATM-346, ATM-347, ATM-349, ATM-350, ATM-351, ATM-352, ATM-353 (2/0 each)` and `ATM-328, ATM-329, ATM-406, ATM-610, ATM-611, SPK-609 (1/0)`. Note the cluster: **8 of the 14 are the June-09 audio/video reopen block — the exact families the operator kept hitting.**
- **Attribution bias:** all 25 recorded reopen events say `by='AI'`; the six operator-driven reopens (mining §4) left no events `[=M6]`.
- **Unguarded done-claims:** done-claiming = terminal + `Ready for testing` = **187** distinct ids. Guard registry (`device/rockchip/rk3588/tests/regression_guard/registry.tsv`): 90 rows, 59 distinct lead keys (44 item-shaped), 70 item-shaped ids counting the `finding_ids` alias column. Join: **14 done-claimers guarded** (ATM-018, ATM-310, ATM-341, ATM-343, ATM-348, ATM-354, ATM-435, ATM-470, ATM-654, ATM-655, ATM-756, ATM-789, ATM-799, SPK-512) → **173 (92.5%) unguarded** `[=M7]`. Needle: known-registered ATM-826 resolves through the same join path.
- **The PC-9 parking lot:** 79 items sit in `Ready for testing` (last_modified 07-09→07-20) — 42% of the done-claiming population is design-complete-but-unverified work that reads as "done, awaiting a formality."

Worst offenders (named): ATM-277 (4 reopens, 0 terminal events, status `Reopened`, family guarded but never verdict-GREEN on a shipped artifact); ATM-346 (audio output switcher, Reopened since 06-09, 2/0); ATM-352 (subtitles-on-primary, Reopened, 2/0, re-minted as ATM-793); ATM-351 (4K glitch, Reopened, 2/0, re-minted as ATM-800); ATM-799 (see G6 — terminal WITH evidence that says PENDING-DEVICE).

Closure: SOL-01 (custody triggers at the DB write seam + full-table sweep at build) + SOL-04 (evidence-class-at-closure). POCs exist and are GREEN (65/65, solutions §5.1); per §11.4.205 they carry no force until wired — the wiring is the Phase 5b item.

. G — The unimplemented-gate surface (PC-)

- Named `CM-*` gates in `constitution/Constitution.md` : **413** `[=M12]` .
- Name-literals resolving in the two canonical gate sites (`pre_build_verification.sh` — 1,028 distinct CM-* tokens, a superset including project-local gates — plus `constitution/scripts/gates/` — 25 tokens): 413 − 239 = 174. **Unimplemented by name-literal: 239** (ROOT_CAUSE measured 241 five days ago; SOL-05's instrument measured 234; the deltas are methodological and stated in SOL-05 §1).
- **Measured join-gap (a §11.4.201(7)(b) needle caught it live in this session):** `CM-RELEASE-VERDICT-COVERAGE` appears in the "unimplemented" list, yet its MECHANISM is fully implemented and wired (`scripts/testing/release_verdict_coverage.sh` invoked from `release_tag.sh`) — the gate-name literal simply appears nowhere under `scripts/` (grep across `scripts/` returned zero files; needle: the literal resolves in `Constitution.md`). Therefore 239 is an **upper bound on missing mechanisms and an exact count of missing name→implementation joins.** Both facts matter: the join gap is precisely why a ledger (SOL-05) is needed — without a name→file mapping, "implemented" is not mechanically decidable, and every future census will wobble.
- 85 literal `gate-code = separate work item` deferrals `[=M12]` .
- **The M11 custody seams remain absent today** (six days after their design): no `CM-STATUS-CUSTODY` literal anywhere in `scripts/` , `pre_build_verification.sh` , or `constitution/scripts/gates/` (needle for the same instrument: the `CM-RELEASE-VERDICT-COVERAGE` literal resolves in `Constitution.md` through the same grep path); no custody sweep script; no adoption-ratchet file (`ls` on both candidate paths: absent).

Ranking by load-bearing weight (which of the 239 matter for 0.0.9): the gates guarding first-touch-failing features and the custody chain outrank doc gates by construction. The load-bearing subset: `CM-STATUS-CUSTODY` (G1), `CM-GUARD-VERDICT-MACHINE-WRITTEN` (G6), `CM-RECURRENCE-LINKS-NOT-MINTS` (G4), `CM-GUARD-ASSERTS-REAL-CONDITION` + `CM-METRIC-VALIDATED-AGAINST-DONE` (G5), `CM-AF-RECENT-WORK-VALIDATION-GATE` / `CM-AF-VALIDATION-ARTIFACT-FILE` (§11.4.46 — the post-flash recent-work gate that would have caught 0.0.7's crash-on-boot class), `CM-RISK-ORDERED-VALIDATION-PRIORITY` + `CM-MOST-REOPENED-EXTRA-DEPTH-LIVE-TEST` (§3 ordering). The ~150 propagation/doc/design gates in the list are real debt but not 0.0.9-blocking; SOL-05's ratchet is how the backlog shrinks without a day-one red wall.

. G — Tracker scope-coverage

Last 60 days, main repo: 1,173 commits; 289 (24.6%) carry an `ATM-|SPK-|FIND-` id in the subject (needle: ATM-808 resolves). Roughly three quarters of commit volume is invisible to every id-keyed custody, guard, and coverage join in this audit — the helix_code analogue (corpus 3: ~80% untracked) reproduced at home. Honest bound: subject-line ids are a proxy; commits whose bodies carry ids or that are legitimately id-free (merges, doc regen) inflate the untracked share — SOL-08's walker classifies properly; this number sizes the hole, it does not adjudicate individual commits.

. G — Operator-experienced recurrences the tracker cannot see (PC-)

Cross-referencing `OPERATOR_COMPLAINT_MINING.md` §4/§6 against today's DB:

Recurrence the operator hit	Tracker state today	Custody state
Brightness sliders ineffective (07-16 ATM-788 → 07-21 **ATM-812, same defect, new id**)	Both open (`Queued`); ATM-812's body/description references ATM-788 **zero times** (query returned 0)	**Zero guards** for either id (registry hits = 0/0) — the only complaint family with NO guard coverage at all
Subtitles on primary display (05-29 → ATM-352 → ATM-239 → 07-16 **ATM-793 new id**)	ATM-352 `Reopened` , ATM-239 `In testing` , ATM-793 `Queued` ; ATM-793 references ATM-352 **zero times**	Family guarded (11 registry hits on ATM-793) but no candidate verdict
4K local glitch (05-28 → ATM-351 → 07-16 **ATM-800 new id**, "mechanism unconfirmed")	ATM-351 `Reopened` , ATM-800 `Queued` ; no link	1-2 registry hits; no candidate verdict
Watch-here / 2nd-display routing (ATM-277 ×4 → 07-16 still broken → 07-21 ATM-809/ ATM-814)	ATM-277 `Reopened` ; ATM-809 `In progress` ; ATM-814 `Queued`	7 registry hits on ATM-277 + smoke guard; no candidate verdict
Audio-switch dialog hang (07-16 ISSUE → 13.6 h REMINDER re-raise → ATM-789)	ATM-789 `Ready for testing`	Guarded (3 hits) + in the guarded∩done set — needs the candidate verdict
exFAT USB regression (ATM-797, title says REGRESSION)	`Queued`	2 registry hits

The `reopens_count` signal is blind to every row above (the re-mints never incremented the originals), and `duplicate-of` has been used 5 times in the project's entire history — intake dedup (SOL-07 / §11.4.214) is unwired in practice. Per §11.4.214's open-target rule, all three re-mint pairs take **LINK-only** (every target is currently open — no reopen event is owed).

NC-1..NC-3 (mining §7) remain unclosed by any SOL: external-state drift on an unchanged artifact (YouTube cert died with zero code change) needs a SCHEDULED standing probe, not a seam gate; complaint-intake latency (13.6 h re-raise) needs the §11.4.210 hook path exercised; no operator-checkable liveness surface exists. These are carried into §3 as punch items, not silently absorbed.

. G — The measurement-layer FAIL/PASS-bluff surface (PC-)

Census re-confirmed: 148 pipefail scripts under `scripts/` + `device/rockchip/rk3588/tests/`; **60 files contain piped `grep -q` under pipefail; 413 total sites** (baseline said 414 — the −1 delta is real drift or pattern-boundary, either way within noise; the SIGPIPE hazard itself was measured 400/400 false-absent at 2 MB in SOL-03's POC). Release-gating classification (measured per file): `pre_build_verification.sh` **118 sites** (the single most load-bearing concentration — a false-FAIL here blocks 0.0.9 spuriously, a false-PASS ships it broken); `build.sh` 16; `regression_guard_suite.sh` 4; `commit_all.sh` 4; `release_tag.sh` 1; `release_verdict_coverage.sh` 0 (clean); `git_hooks/pre-commit` 0 (clean). Priority within the 413: the 143 sites in those six release-path files first; the long tail (flash.sh, helpers) after.

. G — Cross-layer verification gaps (PC- / § . .) — the "bounces on first touch" set

- **82 PENDING-BUILD claim-moves** in the last 60 days of git history (ROOT_CAUSE's 2-month window measured 69; the count has grown).
- **Terminal-event evidence classes:** of the 51 terminal events carrying evidence, **30 cite source-ish artifacts** (research docs, source diffs, gate names) vs 21 runtime-ish (qa-results/recordings/device paths). The discriminator says the 30 are the bounce candidates.

- **Live worst offender:** `ATM-799` (audio jack ES8388 switch) — history row: `Fixed` 2026-07-18, evidence ending in **"(PENDING-DEVICE)"**. A terminal status whose own evidence string states the device layer was never exercised. It is guarded (3 registry rows) — the guard has simply never emitted a verdict on a shipped artifact.
- **The verdict store vs the candidate:** `qa-results/regression_guard/` holds exactly ONE fingerprint — `verdict_latest_93f4f1fd51859f45.tsv`, dated **2026-07-19**, i.e. before the 0.0.8 build (`.last_build_timestamp` = 2026-07-21 02:44:28). Content: 74 guard rows → **24 PASS / 26 FAIL / 24 SKIP**. Meaning today: (a) 26 registered guards were FAILING at the last measured run and nothing in the record shows them turning; (b) the 0.0.8 artifact — the one QA tested — has **zero** verdicts; (c) the release seam, correctly, would refuse a 0.0.9 tag right now for full-registry PENDING. The seam works; the pipeline has never fed it a candidate run. §11.4.135's REQUIRES-REBUILD-PENDING semantics make this honest on an intermediate artifact and blocking on the candidate — the blocking half has never been exercised end-to-end (ROOT_CAUSE §8's last UNCONFIRMED, still true).
- **Recent closures (since 07-16), individually:** ATM-785 (runtime-ish, GREEN 33/33 host-side — sound), ATM-803 (RED+GREEN+negative-control host-side — sound), ATM-815 (Nova retirement, doc evidence — Task, acceptable class), **ATM-799 (PENDING-DEVICE — the live PC-3 instance)**. 1 of 4 recent closures is a device-feature closed at source-green.

2.7 G7 — Danger zones for the 0.0.9 QA session

Base rates: 0.0.7 = crash-loop on first boot (0 minutes to first defect); 0.0.8 = 12 minutes to first defect (YouTube-on-2nd-display, sound-no-picture); ~90% of class-A complaints land within hours of a flash. 61 open critical-severity items today. The ranked family list is §3's spine. One structural risk multiplier stands out: **ATM-816 (`In progress`, Critical) is an in-flight REFACTOR of the video-reproduction path to a fullscreen-mirroring middle-solution** — the single highest-blast-radius change targeted at 0.0.9, landing in exactly the family that broke first touch in BOTH prior releases, while ATM-819 (`In progress`) records that the family's full-automation pipeline itself reported GREEN-while-broken. If 0.0.9 ships ATM-816 without the candidate verdict run + smoke-first ordering, the 12-minute pattern repeats deterministically.

. The punch-list — top by first-touch-QA risk (the Phase b execution order)

Each row: what QA will hit → the mechanism that makes it impossible-or-caught → gating. "NOW" = autonomously doable in this checkout today; gate legend in §4.

#	Punch item	Why it is the risk (measured)	Mechanism (SOL)	Gating
P1	**Run the full regression-guard suite on the 0.0.9 candidate and mint the verdict file; triage the 26 standing FAILs from the 07-19 snapshot first**	Zero verdicts exist for any shipped artifact; 26 guards FAILING at last run; the release seam has never been fed a candidate; both prior releases failed ≤12 min in guarded families	SOL-02 (seam already wired — this is its first real execution) + §11.4.130/132 ordering	BUILD + DEVICE (D1 `998f-d36615e99484`; D2 unavailabl per the durable record)
P2	**Wrong-display video routing family under the ATM-816 refactor: RED-baseline the refactor against `smoke_first_touch.sh` + the FIND-1/ATM-277/ ATM-819 guards BEFORE it merges; smoke runs FIRST post-flash**	≥9 ids; first-touch failure in BOTH prior releases; the refactor is the largest in-flight blast radius; the family's own automation is the recorded bluff (ATM-819)	SOL-04 (class-matched evidence: pixels/liveness, never greps) + existing smoke guard + §11.4.132	Source-side NOW; verification BUILD + DEVICE

#	Punch item	Why it is the risk (measured)	Mechanism (SOL)	Gating
P3	**Wire SOL-01 custody (DB triggers + full-table sweep) so no new terminal status can land without an event + class-matched evidence; honesty-sweep the existing 52 zero-history terminal items and the ATM-799 class** (reclassify or attach evidence via the sanctioned tool — never raw SQL)	48% zero-history; 14 impossible sequences; ATM-799 terminal with "(PENDING-DEVICE)" evidence; the 6/6-broken sample came from exactly this population	SOL-01 + SOL-04	NOW (DB writes via sanctione tool; the pre-build sweep wirin coordinates with Phase 4 — contention path `pre_build_verification.sh`
P4	**Intake dedup + link backfill: land the SOL-07 intake check; LINK the three measured re-mint pairs (ATM-812→ATM-788, ATM-793→ATM-352, ATM-800→ATM-351); backfill the six operator-experienced reopens as User-attributed history events**	The fragility signal is blind exactly on the repeat offenders; 25/25 events "AI"; `duplicate-of` used 5× ever; §11.4.214 open-target rule ⇒ LINK-only (no reopen minting owed)	SOL-07 (+ SOL-01 intake half)	NOW

#	Punch item	Why it is the risk (measured)	Mechanism (SOL)	Gating
P5	**Audio-switching family: verify ATM-799 on the candidate FIRST (its guard exists and has never run on a shipped artifact); ATM-789 + ATM-346 through their guards at §11.4.189 extra depth**	Audio = operator top priority (§11.4.72); ATM-346 `Reopened` since 06-09; ATM-799 is the live source-green closure; dialog-hang needed a 13.6 h re-raise to even get tracked	SOL-02 execution + §11.4.189	BUILD + DEVICE
P6	**Author the missing brightness guard (ATM-788/ATM-812) — the only complaint family with ZERO guard coverage — RED on the current artifact per §11.4.115**	Operator hit it twice, 5 days apart, two ids, no guard, no link; a guaranteed third hit otherwise	§11.4.115 RED-first + registry row	Authoring NOW; RED validatic DEVICE

#	Punch item	Why it is the risk (measured)	Mechanism (SOL)	Gating
P7	**Needle the release path: apply SOL-03's needled-measurement primitive to the 143 piped-`grep -q` sites in the six release-gating files (pre_build 118, build.sh 16, suite 4, commit_all 4, release_tag 1)**	A false-FAIL blocks 0.0.9 spuriously (operator trust damage per §11.4.201(1)); a false-PASS ships it broken; 400/400 false-absent reproduction at 2 MB is on file	SOL-03	NOW for the library + non-contended files; `pre_build_verification.sh` edits coordinate with Phase 4
P8	**First-boot/crash-loop + exFAT + touch-degradation smoke: extend `smoke_first_touch.sh` (or register siblings) for boot-to-launcher-stable, exFAT mount (ATM-797), sustained-touch (ATM-818), CEC power (ATM-817)**	0.0.7's 0-minute crash-loop had no first-boot smoke; ATM-797 is a titled REGRESSION; both are cheap deterministic probes	§11.4.135 registry rows + §11.4.46 recent-work gate	Authoring NOW; validation BUILD + DEVICE

#	Punch item	Why it is the risk (measured)	Mechanism (SOL)	Gating
P9	**Land the SOL-05 gate ledger + ratchet (name→implementing-file mapping; monotone-decrease baseline over the 239)** — including the measured join-gap class (mechanism-implemented-without-literal)	239 name-unimplemented; the custody seams themselves sat prose for 6 days; without the ledger every future census wobbles and new gates join the 239	SOL-05	NOW
P10	**Stand up the NC-1 scheduled standing probe (YouTube/device-cert reachability on a timer, not at seams) + the SOL-09 detection-pressure scheduler so guards run between releases, not only at them**	A shipped feature died with NO code change (05-29 cert drift); corpus-3 recurrence latency = the next sweep, not the next use	SOL-09 + new scheduled probe (NC-1 is outside every current SOL — honest gap being closed, not silently absorbed)	Scheduler NOW; live probe DEVICE + network

Deliberately-surfaced operator decisions (never autonomous, §11.4.122/§11.4.66): (a) the §11.4.140/§11.4.141 anchor-number collision (fleet-wide citation change — already flagged in the constitution status log); (b) the semantics of the 79-item `Ready for testing` parking lot (PC-9 — attaching an evidence precondition per state is a governance change); (c) SOL-05 ratchet + §11.4.224 coverage-floor brownfield adoption mode; (d) triage priority of the 26 standing guard FAILs if any prove to be won't-fix-class. Phase 5b should batch these into ONE §11.4.66 interactive round rather than four blocking stops.

. Doability matrix (what gates what)

Gate	Items	Notes
NOW (autonomous, this checkout, no build/device)	P3 (DB custody wiring + honesty sweep), P4 (dedup + links + User-reopen backfill), P6/P8 authoring halves, P7 library + non-contended files, P9 ledger+ratchet, P10 scheduler half; drafting the P2 RED baselines	DB writes go through the sanctioned mutation tool only (raw SQL is the PC-1 bypass being closed); constitution-submodule edits follow §11.4.26
Phase-4 (`anti_bluff` submodule) gated	The helper layer P3/P7 call into (`ab_*` evidence/needle helpers), and any edit to `pre_build_verification.sh` / `meta_test_false_positive_proof.sh` (Phase 4's declared contention paths)	Submodule not present yet (probe: `constitution/submodules/` = clickup_sync, continuum, session_orchestrator, token_optimizer). Everything in the NOW column is structured to not block on it
BUILD gated	P1, P5, P8-validation, P2-verification (need the 0.0.9 candidate artifact)	One containerized build at a time (§12.9)
DEVICE gated	P1, P2, P5, P6-RED, P8, P10-live	D1 `998fd36615e99484` only; D2 `66ff9c4f51f00ee7` unavailable per the durable session record — §11.4.3 topology-honest single-device verdicts, never a faked dual-device PASS

Gate	Items	Notes
5 **OPERATOR gated**	The four decisions listed in §3	Batch into one §11.4.66 round

. UNKNOWN / honest-boundary register

- **UNKNOWN: the true reopen rate.** Still an undercount (G4 proves the re-mint channel is live); a corrected figure requires the P4 link backfill to land first.
- **UNKNOWN: how many of the 239 name-unimplemented gates have mechanism-without-literal implementations** (the CM-RELEASE-VERDICT-COVERAGE class). ≥1 proven; a full resolution IS the P9 ledger — hand-auditing 239 names here would reproduce the census wobble the ledger exists to end.
- **UNKNOWN: the verdict-snapshot device identity.** `verdict_latest_93f4f1fd51859f45.tsv` carries `device_serial 93f4f1fd51859f45`, which matches neither durable-record serial (D1 `998fd36615e99484`, D2 `66ff9c4f51f00ee7`). Phase 5b must resolve which physical device produced it before treating its 24 PASSes as evidence for anything (§11.4.200's verify-the-intended-target class).
- **UNCONFIRMED: that the 26 guard FAILs of 07-19 are all still-live defects** — some may have been fixed by 0.0.8 work; only the P1 candidate run decides (§11.4.7 — same-conditions evidence, not inference).
- **Boundary:** commit-subject id-bearing (G3) is a proxy for "tracked"; the audit sizes the hole, SOL-08 adjudicates it. The G5 site count is syntactic exposure, not proven-broken — the 400/400 SIGPIPE reproduction proves the hazard class, not each site.
- **Boundary:** this audit proves gaps EXIST and are closable; it does not certify that closing them yields zero QA findings (§0). Discovery pressure (§11.4.118) beyond the known families remains Phase 5b/QA's job — the enumerated-coverage claim, not the clean-bill claim.

. Anti-bluff certification

Read-only on the project: nothing was written except this file (`remediation/` created for it). The DB was opened `-readonly` for every query. No governance file, gate site, test, or Phase-4 path was touched; no commit, no push. Every count above was produced this session by the commands in Appendix A; every reported absence ran a control needle through the same instrument path first, and one needle FAILED usefully (§2.2's join-gap — reported as a finding, not absorbed). `find -newermt` was not used anywhere (known-blind on this host). Figures reproduced from the input corpus are cited `[=M#]`; live deltas from the baselines (413-vs-414 sites, 239-vs-241 gates, 82-vs-69 PENDING-BUILD moves) are stated, not smoothed.

Appendix A — measurement commands (all run – – , repo root `/mnt/track1/atmosphere-t1`)

Ref	Command	Needle
A1	`sqlite3 -readonly docs/workable_items.db "SELECT COUNT(DISTINCT atm_id) FROM items"` → 619	non-zero corpus
A2	terminal: `...WHERE status IN ('Fixed (→ Fixed.md)','Implemented (→ Fixed.md)','Completed (→ Fixed.md)')` → 108; minus `atm_id IN (SELECT atm_id FROM item_history)` → **52**	`SELECT COUNT(*) FROM item_history` → 676
A3	`SELECT event_type, COUNT(*) FROM item_history GROUP BY event_type` → Updated 288 / Opened 287 / Completed 25 / Obsolete 25 / Reopened 25 / Fixed 24 / Implemented 2	—
A4	Reopened-status minus Reopened-event ids → **13**; `SELECT "by", COUNT(*) ... event_type='Reopened' GROUP BY "by"` → AI 25	—
A5	per-id `SUM(Reopened), SUM(terminal)` HAVING r>0 AND t=0 → **14 ids** (ATM-277 4/0 top)	second rows return 2/0 distinct ids
A6	done-claiming 4-status query → **187**; registry keys `awk -F'\t' '!/^#/&&NF>1{print $1}' registry.tsv | sort -u` → 59 (44 item-shaped); + col7 aliases → 70; `comm -12` join → **14** guarded	ATM-826 resolves through the join
A7	registry rows `awk NF>1` count → 90	header schema read via `cat -A`
A8	verdict store `ls qa-results/regression_guard/` → 1 fingerprint (20260719T204913Z / 93f4f1fd51859f45); rows `awk '$1=="row"'` → 74; verdict col → 24 PASS / 26 FAIL / 24 SKIP	`.last_build_timestamp` → 1784583868 = 2026-07-21 02:44 (+05)

Ref	Command	Needle
A9	seam wiring: `grep -n regression_guard|verdict scripts/testing/release_tag.sh` → invocation at :125-135; suite PENDING semantics `grep -nE 'exit 3|PENDING' regression_guard_suite.sh` → exit 3 block at :577-586	known token `OWNED_SUBMODULES` → 5 hits
A10	gates: `grep -oE 'CM-[A-Z0-9-]+' constitution/Constitution.md | sort -u` → 413; same over `pre_build_verification.sh` → 1,028; over `constitution/scripts/gates/` → 25; `comm -23` → **239**	known-implemented in-prebuild resolves; `CM-STATUS-CUSTODY` in unimpl list = 1
A11	join-gap: `grep -rln CM-RELEASE-VERDICT-COVERAGE scripts/` → **0 files** while `scripts/testing/release_verdict_coverage.sh` exists + is invoked	literal resolves in `Constitution.md` via same grep form
A12	M11: `grep -rln CM-STATUS-CUSTODY scripts/ pre_build_verification.sh constitution/scripts/gates/` → 0; `ls *ratchet*` both paths → absent	A11's needle (same instrument, known-present literal)
A13	deferrals: `grep -c "gate-code = separate work item" constitution/Constitution.md` → 85	—
A14	pipefail census: 148 pipefail files; `xargs grep -lE '| *grep -q'` → **60 files**; `grep -cE` sum → **413 sites**; per-file: pre_build 118 / build.sh 16 / suite 4 / commit_all 4 / release_tag 1 / release_verdict_coverage 0 / pre-commit 0	first candidate files listed (flash.sh, ...)

Ref	Command	Needle
A15	scope: `git log --since=2026-05-24 --oneline | wc -l` → 1,173; subjects matching `(ATM|SPK|FIND)-[0-9]+` → **289**	ATM-808 resolves
A16	`git log --since=2026-05-24 --pretty=%s%b | grep -ciE 'PENDING-BUILD'` → **82**	—
A17	recent terminal closures since 07-16 → 4 ids; their `item_history.evidence_path` printed verbatim (ATM-799 ends "(PENDING-DEVICE)")	—
A18	terminal-event evidence classes (qa-results/recording/wav/mp4/device ⇒ runtime-ish) → 21 runtime-ish / 30 other; 51/51 non-empty	—
A19	`Ready for testing` → 79 ids, last_modified 2026-07-09..2026-07-20; open critical → 61	—
A20	family links: ATM-812 body/description LIKE '%ATM-788%' → 0; ATM-793 LIKE '%ATM-352%' → 0; `obsolete_details` reasons → duplicate-of 5	non-zero on other reason values proves query path
A21	registry hits per family id (`grep -c <id> registry.tsv`): ATM-788 0, ATM-812 0, ATM-277 7, ATM-793 11, ATM-346 5, ATM-789 3, ATM-351 1, ATM-800 2, ATM-809 1, ATM-814 1, ATM-797 2	ATM-819 → 4
A22	`smoke_first_touch.sh` present (48,965 B, exec) at `device/rockchip/rk3588/tests/` , referenced from `test_all_fixes.sh` , registered as ATM-826 row 1 of registry	—

Ref	Command	Needle
A23	Phase-4 probe: `ls constitution/submodules/` → clickup_sync, continuum, session_orchestrator, token_optimizer (no `anti_bluff`)	dir listing non-empty

Instrument notes for reproducers: the host `grep` is a different regex engine than habitual (`\|` is a literal under `-E`) — all alternations above use unescaped `|` ; never read a pipeline's exit through a formatter; `find -newermt` is blind on this host; the A6 join is id-shaped-keys only — 15 non-item registry keys (FY, EM, FIND-*, ...) join nothing by construction and are part of G2's ledger work.